

Relational σ -Additive Probability on Orthomodular Lattices: A Working Problem-Set

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Abstract

A self-contained kit for attacking the descent-axis open problem under the *Reading-1* (relational = no hidden realisation) framing. Everything needed to sit down with pen and paper is defined inline: orthomodular lattices and the orthogonal/meet-zero gap; states, dispersion-free states, concreteness; contextuality and the Fine equivalence; the centre and the Stone/Loomis–Sikorski correspondence; and the construction $\prod_n \text{MO}_2$. The open target is stated formally (§7), reduced to a single existence question, and broken into concrete sub-tasks (§8) carrying the constraints established so far (the Wright bound; the off-centre requirement from the CE-routing analysis). Companion narrative: `oml_onboarding.tex`, `oml_onboarding.md`, `reading1_prize_reduction.md`, `sigma_essential_nonemptiness_finding.md`. This document is presentation, not proof: open questions are flagged as open; included proofs are short and self-verifying or cited.

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1 Orthomodular lattices and the orthogonal/meet-zero gap

Definition 1.1 (Orthomodular lattice). An *orthomodular lattice* (OML) is a bounded lattice $(L, \wedge, \vee, 0, 1)$ with an order-reversing involution $a \mapsto a^\perp$ such that

$$a \vee a^\perp = 1, \quad a \wedge a^\perp = 0, \quad (a^\perp)^\perp = a,$$

and satisfying the *orthomodular law*:

$$a \leq b \implies b = a \vee (b \wedge a^\perp).$$

Write $a \perp b$ (“*a orthogonal b*”) for $a \leq b^\perp$. L is *Boolean* iff it is distributive.

Definition 1.2 (The two relations, and the gap). For $a, b \in L$:

$$a \perp b :\iff a \leq b^\perp \quad (\text{orthogonal}), \quad a \wedge b = 0 \quad (\text{meet-zero}).$$

Always $a \perp b \Rightarrow a \wedge b = 0$. The converse *fails* in a non-Boolean OML; the failure is the source of everything here.

Example 1.3 (MO_2 , the minimal gap). $\text{MO}_2 = \{0, 1, a, a^\perp, b, b^\perp\}$ (the “Chinese lantern”): four atoms, each pair of distinct atoms joining to 1 and meeting to 0. So $a \wedge b = 0$ but $a \not\leq b^\perp$ (as $b^\perp \in \{b^\perp\}$ and $a \neq b^\perp$), i.e. $a \wedge b = 0$ yet $a \not\perp b$. MO_2 is the smallest non-Boolean OML and the unit of every construction below.

Definition 1.4 (Completeness notions). L is σ -complete if every countable family $\{a_n\} \subseteq L$ has a join $\bigvee_n a_n \in L$. L is σ -orthocomplete if this holds for every countable *pairwise-orthogonal* family. (σ -complete $\Rightarrow \sigma$ -orthocomplete; the converse fails on non-Boolean OMLs.)

Definition 1.5 (Centre). $z \in L$ is *central* if it is compatible with every element and $L \cong [0, z] \times [0, z^\perp]$ as OMLs. The *centre* $Z(L) = \{\text{central elements}\}$ is a Boolean subalgebra; L is *irreducible* if $Z(L) = \{0, 1\}$.

2 States, dispersion-free states, concreteness

Definition 2.1 (State). A *state* on L is $s: L \rightarrow [0, 1]$ with $s(1) = 1$ and finite additivity on orthogonal pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$. It is σ -additive if $s(\bigvee_n a_n) = \sum_n s(a_n)$ for countable pairwise-orthogonal $\{a_n\}$ (join taken in L). A state is *dispersion-free* (2-valued) if $s(L) \subseteq \{0, 1\}$.

Definition 2.2 (Concrete logic). A set S of states is *order-determining* if $(\forall s \in S : s(a) \leq s(b)) \Rightarrow a \leq b$. L is *concrete* (equivalently *set-representable*) iff it admits an order-determining set of *dispersion-free* states (Gudder [3]). Equivalently, L embeds into some $(\mathcal{P}(X), \subseteq, X \setminus (-))$ preserving existing orthogonal joins.

Remark 2.3 (Separating vs. spanning — the load-bearing distinction). Concreteness asks the dispersion-free states to *separate* the order. It does *not* ask them to *span* the state space (i.e. that every state be a mixture of them). The whole problem lives in this gap; keep it in view.

Fact 2.4 (MO_2 has no dispersion-free homomorphism, but is concrete). MO_2 has *exactly four* order-determining dispersion-free states (each sending one atom-pair to 0 and the other to 1), so MO_2 is concrete. None is a lattice homomorphism: for the state s with $s(a) = s(b) = 0$ one has $s(a \vee b) = s(1) = 1 \neq 0 = \max(s(a), s(b))$ (equivalently $1 > 0 = s(a) + s(b)$, so s is not subadditive). Concrete uses states, not homomorphisms.

3 Contextuality and the Fine equivalence

The Reading-1 reading of “relational” — the measure carries *no hidden realisation* — is made precise by contextuality. Fix a concrete OML L with order-determining dispersion-free set S_{df} .

Definition 3.1 (Contextual state). A σ -additive state w on L is *non-contextual* if it lies in the closed convex hull $\overline{\text{conv}}(S_{\text{df}})$ of the dispersion-free states (in the topology of pointwise convergence on L); otherwise *contextual*.

Theorem 3.2 (Fine, in lattice form). For a concrete L , the following are equivalent for a σ -additive state w :

- (i) w is non-contextual ($w \in \overline{\text{conv}}(S_{\text{df}})$);

- (ii) $w = \int_{S_{\text{df}}} s d\mu(s)$ for a probability measure μ on S_{df} (a global hidden joint);
- (iii) the context-wise distributions w induces on the Boolean blocks of L admit a single global joint distribution reproducing all of them.

A dispersion-free state is a deterministic global section, so a global joint and a hull-decomposition are the same object (Fine [2]; Abramsky–Brandenburger [1]).

Remark 3.3 (Three point-spaces — do not conflate). The recurring trap. Distinguish:

- (A) the *external* points (e.g. Hilbert rays of H for $L(H)$): an auxiliary space the measure could be reduced to (Gleason’s ρ in $s = \text{tr}(\rho \cdot)$). Reading-1 *forbids* reduction to these.
- (B) the *canonical dual* points $P(A) = \{\uparrow p : p \in L\}$ (principal filters — one per *element*): built from L , hence *permitted*.
- (C) the *dispersion-free states* S_{df} (one per consistent global *valuation*): the hidden-variable points of Def. 3.1.

“Relational = contextual” is a condition on (C), *not* on (B): a witness is point-rich on $P(A)$ yet has no global S_{df} -joint. (A), (B), (C) are genuinely different objects.

The open target. Relational σ -additive probability = a contextual σ -additive state on a concrete OML. (The formal open problem is §7.)

4 The Boolean engine, and why it does not transfer

Theorem 4.1 (Stone / Loomis–Sikorski). *For a Boolean algebra B with Stone space $\text{St}(B)$: B is σ -complete iff $\text{St}(B)$ is basically disconnected (the closure of every cozero set is open); and a finitely-additive state on B is σ -additive iff, viewed as a measure on $\text{St}(B)$, it concentrates on the principal (isolated/atomic) part. In particular a σ -additive state on $\mathcal{P}(\mathbb{N})$ is atomic: $w = \sum_n w(\{n\})\delta_n$.*

Theorem 4.2 (Pták–Pulmannová [4]). *An OML is Boolean iff it carries a unital set of subadditive states. Hence the Boolean-forcing property is subadditivity, not σ -additivity: σ -additivity alone does not collapse a non-Boolean OML, so non-Boolean σ -additive states genuinely exist.*

Remark 4.3 (The Boolean kill-mechanism, and its Boolean-only hinge). On a Boolean algebra the chain “ σ -additive $\Rightarrow$ concentrates on principal ultrafilters $\Rightarrow$ non-contextual” is forced, because there

$$\text{principal ultrafilter} = \text{atom} = \text{dispersion-free state}.$$

This triple identity is *Boolean-only*. On an OML with centre $Z(L)$: w restricted to $Z(L)$ is a σ -additive Boolean measure, so (by Thm. 4.1) it concentrates on the *central atoms* — but a central atom z subtends a (possibly non-Boolean) block $[0, z]$, and a state with $w(z) = 1$ can be *contextual within* $[0, z]$. So

$$\underbrace{\sigma\text{-add} \Rightarrow \text{concentrate on central atoms}}_{\text{transfers}}, \quad \underbrace{\text{concentrate} \Rightarrow \text{non-contextual}}_{\text{Boolean-only; FAILS}}.$$

For *irreducible* L ($Z(L) = \{0, 1\}$) the centre argument says only $w(1) = 1$ — vacuous. *The Boolean kill-mechanism evaporates exactly off-centre, where a witness would live.* (Routing verdict: `sigma.essential.nonemptiness.finding.md`.)

Remark 4.4 (Why no faithful set representation — the universal floor). A faithful σ -tribe / Loomis–Sikorski representation realises L as a field of sets, preserving $\wedge, \vee$ as $\cap, \cup$; its image is a sublattice of a Boolean algebra, hence distributive, forcing L Boolean. So *no* non-Boolean OML admits one. A relational measure can therefore never arise from a faithful descent to points — the target is necessarily “point-free” in the sense of Rmk. 3.3(A), *not* of (B).

5 The construction $\prod_n \text{MO}_2$, and what it settles

Definition 5.1 ($\prod_n \text{MO}_2$). The full direct product of countably many copies of MO_2 , coordinatewise operations. It is concrete (coordinate states separate), σ -complete (product of finite complete lattices), non-Boolean, infinite. Its centre contains $\prod_n \{0, 1\} \cong \mathcal{P}(\mathbb{N})$.

Proposition 5.2 (The interleaving property ($\star$) is cheap). *There exist p and an infinite orthogonal family $\{a_n\}$ in $\prod_n \text{MO}_2$ with $p \wedge a_n = 0$ and $p \not\leq a_n$ for all n . (Take $a_n = a$ in coordinate n , 0 elsewhere; $b_n = b$ in coordinate n , 0 elsewhere; $p = \bigvee_n b_n = (b, b, b, \dots)$, a genuine orthogonal join. Coordinate n : $a \wedge b = 0$ but $b \not\leq a^\perp$.) So “($\star$) + concrete + σ -complete + non-Boolean” is satisfied by a one-line object: it is not the discriminator.*

Proposition 5.3 ($\prod_n \text{MO}_2$ is not a witness — contextuality is only finitely additive). *Every σ -additive state on $\prod_n \text{MO}_2$ is non-contextual (in $\overline{\text{con}}\overline{\text{v}}(S_{\text{df}})$). Sketch: a contextual state would have to act diffusely on the central $\mathcal{P}(\mathbb{N})$ (a free-ultrafilter/Banach-limit type state); but by Thm. 4.1 a σ -additive state on $\mathcal{P}(\mathbb{N})$ is atomic, hence concentrates on coordinates, hence is a countable mixture of coordinate dispersion-free states. The contextual states exist only in the finitely additive layer; σ -additivity kills them. (This is the CE / Paper-I compactness phenomenon, and the reason $\prod_n \text{MO}_2$ fails: all its infinitary structure is central.)*

6 The bound: Wright’s pentagon

Theorem 6.1 (Wright [5]). *There is a finite, concrete OML — the pentagon (pentagram) logic, a Greechie loop of length five (hence a lattice) — with an order-determining family of dispersion-free states, carrying a state not in their convex hull, i.e. a contextual state.*

Remark 6.2 (What Wright settles, and what it leaves). A finite OML is vacuously σ -complete and its states vacuously σ -additive. So “concrete σ -complete non-Boolean OML with a σ -additive contextual state” is *already inhabited* by Wright — and the general spanning theorem “every state is a dispersion-free mixture” is *false*. What Wright does *not* give: contextuality requiring the *countable* structure. Its contextuality is witnessed by a finite sub-OML (itself). That single missing qualifier — **σ -essential** — is the entire open content.

7 The open target

*The open target. (**σ -essential contextuality**).* Does there exist a concrete, σ -complete, non-Boolean OML L and a σ -additive state w on L such that:

- (1) w is *contextual* — $w \notin \overline{\text{con}}\overline{\text{v}}(S_{\text{df}})$ (no global hidden joint over the dispersion-free states); and
- (2) the contextuality is *σ -essential* — the restriction of w to *every finite sub-OML* of L is non-contextual (so the obstruction lives only in the countable structure)?

Equivalently: a relational σ -additive probability whose relationality is not already present finitely. **Or:** prove no such (L, w) exists.

Remark 7.1 (Status). **Well-posed, principled, uninhabited.** No witness ($\prod_n \text{MO}_2$ fails by Prop. 5.3; Wright is finite, Rmk. 6.2); no impossibility proof. The CE-routing analysis (Rmk. 4.3) returned *GAP*: CE neither forbids nor produces a witness, but localises both exits (§8).

8 Sub-tasks — the pen-and-paper frontier

Two exits; each is genuine mathematics, not orientation. The constraints below are established (Rmk. 4.3, Prop. 5.3) and prune the search.

Exit A — build a σ -essential witness

Sub-task 1 (The off-centre constraint). Show (or use) that a witness must place its infinite non-Boolean structure *off-centre*: L irreducible, or at least with the relevant infinite blocks non-central. *Why*: if all infinitary structure is central, Stone over the centre (Thm. 4.1) forces atomicity and kills contextuality σ -additively (Prop. 5.3). *Excludes products of finite blocks a priori*.

Sub-task 2 (Pasting / countable colimit of Wright blocks). Construct an infinite concrete σ -complete OML as a countable paste/colimit of Wright-type (pentagon) blocks such that:

- (a) each finite sub-OML is non-contextual (dispersion-free states span its state space);
- (b) the centre stays trivial (or the relevant blocks non-central) — Task 1;
- (c) the countable join forces a contextual σ -additive state (a Kochen–Specker-type obstruction appearing only in the limit).

Concretely: can the Greechie/pasting machinery that yields the finite pentagon be iterated to a σ -orthocomplete limit while keeping the centre trivial and concreteness intact? CE raises no objection (Rmk. 4.3); this is the sanctioned shape. *One such witness settles the descent axis positively*.

Exit B — prove no witness exists

Sub-task 3 (A new finite-witnessing theorem). Prove: every σ -additive contextual state on a concrete σ -complete OML is *already* contextual on some finite sub-OML (equivalently, σ -essential contextuality is impossible). **Constraint**: this *cannot* reuse the $\prod_n \text{MO}_2$ /Stone-over-centre mechanism — it is vacuous off-centre (Rmk. 4.3). A genuinely different argument is required. *Lever*: the Pták–Pulmannová subadditivity boundary (Thm. 4.2); the missing bridge is subadditivity $\leftrightarrow$ contextuality at the σ level. *Such a theorem reinstates the empiricist-underdetermination conclusion as a theorem* (`direction2_gate_finding.md`).

Remark 8.1 (Discipline). Neither exit is decidable by a quick computation; both are Phase-4 mathematics. The recurring failure mode in this material is the clean “ X , therefore Y ” step that needs a qualifier (the Boolean triple identity of Rmk. 4.3 is the paradigm). Verify transfers; do not assume them. A candidate witness is checked by: is its claimed contextual state genuinely outside $\overline{\text{conv}}(S_{\text{df}})$, and is the obstruction genuinely absent from every finite sub-OML?

References

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